THE MASTERMIND COACHING CLASSES

Revision Notes

Class 10 · Science

Biology — Chapters 5 to 8 · NCERT Science — Class 10 · CBSE / RBSE Syllabus 2026-27

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
5	Life Processes	Nutrition, respiration, transport and excretion in plants and animals
6	Control and Coordination	Nervous system, reflex action, brain, plant and animal hormones
7	How do Organisms Reproduce?	Asexual and sexual reproduction in plants and humans
8	Heredity	Inheritance of traits, Mendel's experiments, sex determination

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CHAPTER 05 · BIOLOGY

Life Processes

A living body is never idle: even during sleep it feeds, breathes, circulates fluids and removes wastes. Without these maintenance jobs, the body would break down. This chapter follows four sets of life processes: how organisms obtain and use food (nutrition), how they get energy from it (respiration), how materials move around the body (transportation) and how wastes are removed (excretion).

Big Questions

- How do organisms obtain food and convert it into energy?
- How are materials carried to and from every cell?
- How are metabolic wastes removed from plants and animals?

5.1 What are Life Processes?

Life processes are the basic activities that **maintain** living things. They need energy, raw materials and a mechanism of exchange. Because single cells and even whole bodies are exposed to damage, organisms constantly repair themselves through **nutrition, respiration, transportation and excretion**.

5.2 Nutrition

DEFINITION

Nutrition is the process of obtaining food and using it for energy, growth and repair. **Autotrophs** make their own food from simple inorganic substances; **heterotrophs** depend on other organisms.

▸ Autotrophic nutrition: photosynthesis

FORMULA

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ in the presence of sunlight and chlorophyll.

1 — Absorption of light energy by chlorophyll
Light is trapped by the green pigment in chloroplasts.

2 — Splitting of water
Light energy converts into chemical energy, and water splits into hydrogen and oxygen.

3 — Reduction of carbon dioxide
Carbon dioxide is reduced to carbohydrate; these steps need not take place together.

- **Stomata** are tiny pores on leaves, with **guard cells** that control opening. They let CO_2 in and O_2 out; much water vapour also escapes. Stomata close when guard cells lose water.
- Carbon dioxide is taken from air; water and minerals (nitrogen, phosphorus) come from the soil through roots.
- Starch is the stored product; its presence in leaves is tested with iodine.
- Desert plants take CO_2 at night and prepare an intermediate that is used during the day.

▸ Heterotrophic nutrition

Mode	Description	Examples
Saprophytic	Break down dead and decaying matter outside the body, then absorb it	Fungi: bread mould, yeast, mushrooms
Parasitic	Derive food from a living host without killing it quickly	Cuscuta, tapeworm, lice, ticks
Holozoic	Ingest solid food and digest it internally	Amoeba, humans, dogs

Amoeba engulfs food with pseudopodia into a food vacuole, digests it with enzymes, absorbs it and expels the leftover. **Paramecium** takes food through a fixed spot by cilia movement.

▸ Nutrition in human beings

Part	Function
Mouth	Teeth chew; **salivary amylase** begins digesting starch to sugar; tongue mixes saliva
Oesophagus	Peristaltic movement pushes food to the stomach
Stomach	**HCl** creates acidic medium, kills bacteria and activates **pepsin**, which digests proteins; mucus protects the lining
Small intestine	Longest part. **Bile** from the liver emulsifies fats and makes food alkaline; pancreatic juice has **trypsin** (proteins), **lipase** (fats), amylase; intestinal juice finishes digestion. **Villi** absorb food
Large intestine	Absorbs water; the rest is egested through the anus

- Liver is the largest gland; the pancreas secretes enzymes and insulin.
- Dental caries: sugars and bacteria form acid that dissolves enamel; brushing removes plaque.

5.3 Respiration

DEFINITION

Respiration is the process of breaking down food to release energy, stored in **ATP**. It occurs in the cell, first in cytoplasm (glycolysis, glucose to **pyruvate**) and then in mitochondria (aerobic).

Pathway	Where	Products
Aerobic	Mitochondria, with oxygen	CO_2 + H_2O + **38 ATP** worth of energy
Anaerobic in yeast	Cytoplasm, without oxygen	Ethanol + CO_2 + energy
Anaerobic in muscle	When oxygen is short	**Lactic acid** + energy; build-up causes cramps

FORMULA

Glucose → pyruvate (3C) in cytoplasm; pyruvate → CO_2 + H_2O + energy (aerobic) | pyruvate → ethanol + CO_2 (yeast) | pyruvate → lactic acid (muscle). ATP → ADP + P releases energy for cell activity (about 30.5 kJ per mole).

- **Plants:** gas exchange through stomata (leaves), lenticels (stems) and root surface; plants need less exchange as their demand is lower.

- **Humans:** air passes nostrils → pharynx → larynx → trachea (cartilage rings) → bronchi → bronchioles → **alveoli**, thin-walled balloon-like air sacs with rich blood supply; here O_2 enters blood and CO_2 leaves.
- **Haemoglobin** in red blood cells carries oxygen; CO_2 is carried mostly dissolved in plasma. Lung disease or low haemoglobin cause breathlessness.
- Aquatic animals breathe faster because water holds little dissolved oxygen; fish take it with gills.

5.4 Transportation

▸ In human beings

The human heart is a muscular four-chambered organ: two **atria** (receive) and two **ventricles** (pump; thicker walls). A septum keeps oxygenated and deoxygenated blood separate, which supports the high energy needs of warm-blooded animals (**double circulation**).

Pulmonary circuit
Right ventricle → lungs (CO_2 out, O_2 in) → left atrium.

Systemic circuit
Left ventricle → body → right atrium via vena cava.

Valves
Prevent backflow of blood.

Component	Role
Plasma	Carries food, CO_2, nitrogenous wastes in dissolved form
Red blood cells	Haemoglobin carries O_2
Platelets	Plug leaks by clotting at injured points
Arteries	Thick elastic walls, carry blood **away** from heart under pressure
Veins	Thin walls with valves, carry blood **to** heart
Capillaries	One cell thick; exchange of materials
Lymph	Colourless fluid from tissues; carries digested fat and drains excess fluid

Blood pressure is the force against vessel walls: normal about **120 mm Hg** systolic and **80 mm Hg** diastolic; high blood pressure can rupture vessels.

▸ In plants

Tissue	Carries	Process
Xylem	Water and minerals upward	Root pressure at night; **transpiration pull** during the day pulls the water column
Phloem	Food (sucrose) in both directions	**Translocation**: uses ATP energy; sugar loaded into phloem raises osmotic pressure and water enters

5.5 Excretion

DEFINITION

Excretion is the removal of harmful metabolic wastes from the body.

- **Human excretory system:** a pair of kidneys, two ureters, a urinary bladder and a urethra. Kidneys remove nitrogenous waste (urea, uric acid) from blood.
- **Nephron**, the unit of the kidney: a cluster of capillaries called **glomerulus** inside **Bowman's capsule**, plus a long tubule. Filtration at the glomerulus; useful glucose, amino acids and water are **reabsorbed**; the leftover is urine. A person produces 1.0 to 1.5 L daily.
- **Artificial kidney (haemodialysis)** cleans the blood of patients with kidney failure: blood flows through tubes bounded by a semi-permeable membrane in a dialysing fluid with no nitrogenous waste.
- **Plants:** excess water by transpiration, gases by stomata, wastes stored in leaves that fall, resins and gums in old xylem, and latex.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why is the small intestine long and villi present?	Greater area for digestion and absorption of food.
What is the role of bile?	It makes the food alkaline and emulsifies fats.
Why do muscles cramp after heavy exercise?	Anaerobic respiration forms lactic acid.
How are stomata opened and closed?	Guard cells swell when water enters (open) and shrink when it leaves (close).
Differentiate xylem and phloem.	Xylem carries water and minerals upward; phloem carries food in both directions.
Why is double circulation needed?	Oxygen-rich and oxygen-poor blood are kept separate for efficient supply of energy.
Name the functional unit of the kidney.	Nephron.
Why is air in the lungs not completely emptied?	A residual volume keeps alveoli open, so oxygen supply continues.

★ KEY TAKEAWAYS

- **Photosynthesis:** $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$; stomata control gas exchange.
- Human digestion: **amylase** (starch), **pepsin** (protein), **bile + lipase** (fat); absorption in **villi**.
- Respiration releases energy as **ATP**; anaerobic gives ethanol (yeast) or lactic acid (muscle).
- Heart: **four chambers, double circulation**; arteries away, veins towards the heart.
- Xylem moves water by **transpiration pull**; phloem moves food by **translocation**.
- **Nephron** filters blood; dialysis replaces kidney function.

CHAPTER 06 · BIOLOGY

Control and Coordination

When you pull your hand from a hot pan, notice a smell or decide to walk, many organs act in step. This is coordination. Animals use two systems for it: the fast electrical nervous system and the slower chemical hormones; plants rely on hormones alone. This chapter follows how a stimulus becomes a response and how growth is regulated.

Big Questions

- How does the body detect a stimulus and respond?
- What are the parts and functions of the human brain?
- How do plant and animal hormones coordinate life?

6.1 Animals: The Nervous System

Information from the environment is received by **receptors** in sense organs (ear: sound and balance; eye: light; nose: smell; tongue: taste; skin: touch). The signal travels through the **neuron**, the unit of the nervous system, to the brain or spinal cord, and a response is sent back to **effectors** (muscles and glands).

Part of the neuron	Function
Dendrite	Collects information; the end of the neuron where signals are acquired
Cell body	Information travels as an electrical impulse
Axon	Carries the impulse away to the end
Synapse	Gap between neurons; chemical released converts the impulse to the next neuron

Receptor
Detects the stimulus.

Sensory neuron
Carries impulse to CNS.

Relay neuron (spinal cord / brain)
Processes information.

Motor neuron
Carries message to effector.

Effector
Muscle or gland shows the response.

6.2 Reflex Action

DEFINITION

A **reflex action** is a sudden, automatic response to a stimulus without conscious thought. The pathway is called the **reflex arc**. It is processed in the **spinal cord**, so the response is fast and saves the body from harm.

Feature	Reflex action	Voluntary action
Control	Spinal cord (sometimes brain)	Fore-brain
Thought	Not needed	Involves thought
Speed	Very fast	Slower
Examples	Touching a hot object, blinking, knee jerk, coughing	Writing, talking, walking

6.3 The Human Brain

Part	Main functions
Fore-brain: cerebrum	Thinking, memory, voluntary action, senses; has centres for hearing, smell, sight; hunger centre
Mid-brain	Involuntary actions such as pupil size
Hind-brain: cerebellum	Posture, balance, precision of voluntary movements
Hind-brain: pons	Regulates breathing
Hind-brain: medulla	Involuntary actions: blood pressure, vomiting, salivation

- The brain and spinal cord form the **central nervous system**; nerves outside it form the **peripheral nervous system**.
- The brain is protected by the **skull** (bony box) and **cerebrospinal fluid** in a fluid-filled balloon; the spinal cord by the **vertebral column**.
- Muscles move by changing shape: muscle cells contain special proteins that change shape and length in response to nervous signals.

6.4 Coordination in Plants

Plants have no nervous system or muscles, yet they respond. Some movements are **independent of growth**, as in the touch-me-not (*Mimosa pudica*), which droops when touched because cells quickly lose water and shrink. Other movements are **growth-related** and called **tropisms**.

Tropism	Stimulus	Response
Phototropism	Light	Shoots bend towards light; roots bend away
Geotropism	Gravity	Roots grow down (positive), shoots up (negative)
Hydrotropism	Water	Roots grow towards water
Chemotropism	Chemicals	Pollen tube grows towards the ovule

Plant hormone	Function
Auxin	Made at the shoot tip; helps cells grow longer; on the shaded side of the shoot it promotes growth, so the shoot bends towards light
Gibberellins	Help stem growth

Plant hormone	Function
Cytokinins	Promote cell division; present in fruits and seeds
Abscisic acid	Inhibits growth; causes wilting of leaves, closing of stomata

6.5 Hormones in Animals

Chemical messengers released by **endocrine glands** directly into the blood act on target organs; they are needed in small amounts and work slowly but for longer than nerve signals. The level of a hormone is controlled by **feedback** (e.g., blood sugar).

Gland	Hormone	Function / disorder
Pituitary	Growth hormone	Controls growth; lack gives dwarfism, excess gives gigantism; master gland
Thyroid	**Thyroxine**	Controls metabolism; needs **iodine**; deficiency causes **goitre**
Pancreas	**Insulin**	Lowers blood sugar; lack leads to **diabetes**
Adrenal	**Adrenaline**	Prepares for emergency: heartbeat and breathing quicken
Testes	Testosterone	Male puberty changes
Ovary	Oestrogen	Female puberty changes

IMPORTANT

Iodised salt is recommended because iodine is essential for thyroxine; the pancreas works as both a digestive and an endocrine gland.

6.6 Worked Examples

Trace the pathway when a hot object is touched: thermal receptors in skin → sensory neuron → spinal cord (relay neuron) → motor neuron → arm muscles contract and the hand is withdrawn.

Why is a reflex arc useful? Thinking time in the brain would be too long; in the spinal cord the path is short and fast.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is a synapse?	The gap between the end of one neuron and the dendrite of the next, crossed by chemicals.
Which part controls posture and balance?	Cerebellum.
Why is the response of a plant to touch different from a growth movement?	The movement of touch-me-not occurs through changes in water in cells, not growth.
Name the hormone that lowers blood sugar.	Insulin.
Why is iodised salt recommended?	Iodine is needed to make thyroxine and prevents goitre.
Name a hormone released in emergency.	Adrenaline.

Likely question	Model answer
How does auxin help the shoot bend towards light?	It collects on the shaded side and makes those cells grow longer.
Differentiate nervous and hormonal control.	Nerves: fast, electrical, short-lived. Hormones: slow, chemical, long-lasting.

★ KEY TAKEAWAYS

- Stimulus → receptor → **sensory neuron** → CNS → **motor neuron** → effector.
- **Reflex arcs** run through the spinal cord for quick protection.
- Brain parts: **cerebrum** (thought), **cerebellum** (balance), **medulla** (involuntary).
- Plants respond by **tropisms** controlled by hormones: auxin, gibberellin, cytokinin, abscisic acid.
- Animal hormones: **thyroxine, insulin, adrenaline, growth hormone**.

CHAPTER 07 · BIOLOGY

How do Organisms Reproduce?

*Reproduction does not keep an individual alive, but it keeps the **species** alive. Organisms make copies of themselves using DNA copying and a little variation, so that populations survive environmental changes. This chapter compares asexual and sexual reproduction, follows flowering plants from pollination to seed and explains the human reproductive systems.*

Big Questions

- Why is variation useful for a species?
- How do plants and animals reproduce with one or two parents?
- How does the human reproductive system work?

7.1 DNA Copying and Variation

Cells copy their **DNA** and create extra cell machinery before dividing, so daughter cells carry the same blueprint. The copying is never perfectly accurate, so **variations** appear. Variations help a population: some individuals survive when conditions change, such as heat or a new disease. Bacteria reproduce rapidly; a heat-resistant variant survives a heat wave.

7.2 Modes of Asexual Reproduction

Mode	Description	Examples
Fission	Cell divides into two (binary) or many (multiple) parts	**Amoeba** (binary, any plane), **Leishmania** (binary, fixed plane), **Plasmodium** (multiple)
Fragmentation	Body breaks into pieces that grow into new individuals	*Spirogyra*
Regeneration	A damaged part regrows from specialised cells	*Hydra*, *Planaria*
Budding	A bud grows from the parent and detaches	*Hydra*, yeast
Vegetative propagation	New plants from roots, stem, leaves	Potato (eyes), *Bryophyllum* (leaf notches), rose, sugar cane, banana, jasmine
Spore formation	Tiny spores in sporangia germinate in favourable conditions	*Rhizopus* (bread mould)

- **Tissue culture** grows new plants from a small piece of tissue in a nutrient medium; used for orchids, ornamental plants and seedless fruit.
- Advantages of vegetative propagation: plants raised produce flowers and fruits earlier, banana and seedless plants can be grown, and genetic features stay the same.
- Asexual reproduction needs one parent; offspring are identical (clones) with very little variation.

7.3 Sexual Reproduction

Two parents contribute **gametes** (male and female sex cells) which fuse to form a **zygote** by **fertilisation**. DNA is combined from two sources, so variation is greater. Gametes are produced by a special cell division that halves the number of chromosomes, so the offspring has the normal number again.

▸ Sexual reproduction in flowering plants

Flower part	Function
Sepals	Protect the bud
Petals	Attract insects
Stamen (anther + filament)	Male part; anther makes **pollen**
Carpel (stigma, style, ovary)	Female part; ovary holds **ovules** with the egg cell

Pollination
Transfer of pollen from anther to stigma by wind, water or insects. **Self-pollination** is within the same flower; **cross-pollination** is between flowers of different plants.

Pollen tube
Pollen grain on the stigma grows a tube through the style to the ovary.

Fertilisation
Male gamete fuses with the egg in the ovule and forms the zygote.

After fertilisation
Zygote becomes the embryo, ovule becomes the **seed**, ovary becomes the **fruit**, other flower parts fall off. Seeds germinate to give new plants.

A unisexual flower has either stamens or carpels (papaya, watermelon); a bisexual flower has both (hibiscus, mustard).

7.4 Reproduction in Human Beings

Humans are sexually reproducing, unisexual and show **internal fertilisation**. At **puberty** (about 10 to 12 years) body changes take place: a boy's voice deepens and facial hair grows; a girl's breasts develop and menstruation begins.

Male system	Function
Testes (in the scrotum)	Make **sperm** and **testosterone**; scrotum keeps them cooler than body temperature, which sperm production needs
Vas deferens	Carries sperm
Seminal vesicles, prostate gland	Add fluid to make **semen** and nourish sperm
Urethra	Common passage for semen and urine

Female system	Function
Ovaries	Make **eggs** and hormones; an egg is released about every month
Oviducts (fallopian tubes)	Carry the egg; fertilisation takes place here

Female system	Function
Uterus	Where the embryo is implanted and develops
Cervix and vagina	Opening into the uterus; birth canal

Fertilisation
Sperm fuses with egg in the oviduct to form the zygote.

Implantation
The embryo implants in the thickened, blood-rich uterine lining.

Placenta
A disc-like structure connecting embryo and uterine wall; provides glucose, oxygen and removes wastes (about nine months, **gestation**).

Menstruation
If no fertilisation, the lining and blood are shed through the vagina about every 28 days (2 to 8 days).

IMPORTANT

Egg and sperm are the **sex cells**. The ovary, testis, and the hormones testosterone and oestrogen differ between sexes, but both parents contribute equally to the genetic make-up of the child.

7.5 Worked Examples

Why does Leishmania divide at a fixed plane while Amoeba does not? Leishmania has a whip-like structure at one end, so binary fission follows a definite orientation.

Why do bread slices left in a humid place grow a cotton-like mould? *Rhizopus* spores in air land on bread and germinate; blobs of black sporangia form on the hyphae.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why is variation useful for a species?	It helps some members survive when the environment changes.
Differentiate binary fission and multiple fission.	Binary: parent splits into two; multiple: into many daughter cells (Plasmodium).
How does Hydra reproduce by budding?	A bud develops as an outgrowth due to repeated cell division at one site and detaches as a tiny Hydra.
What is the role of the pollen tube?	It carries the male gamete to the ovule.
Why are testes in the scrotum?	Sperm formation needs a temperature lower than body temperature.
What is the placenta?	A tissue that passes nutrients from mother to embryo and wastes back.
What happens to the ovary after fertilisation?	It ripens into the fruit and the ovule becomes the seed.
How is tissue culture useful?	Many identical disease-free plants are raised from a small tissue.

★ KEY TAKEAWAYS

- Reproduction copies **DNA** with small errors, producing the **variation** needed by a species.
- Asexual: **fission, fragmentation, regeneration, budding, vegetative propagation, spores** (one parent, clones).
- Flower: **pollination → pollen tube → fertilisation → seed and fruit**.
- Human male: **testes, vas deferens, urethra**; female: **ovary, oviduct, uterus**.
- Fertilised egg → zygote → embryo implants in uterus, fed through the **placenta**.

CHAPTER 08 · BIOLOGY

Heredity

Children resemble their parents, yet are not exact copies, and brothers and sisters differ from each other. Heredity is the passing of traits from parents to offspring; variation is the difference between individuals. This chapter shows how Mendel, working with garden peas, discovered the rules of inheritance and how the sex of a child is decided.

Big Questions

- How are traits passed from one generation to the next?
- What did Mendel's pea experiments reveal?
- What decides whether a child is a boy or a girl?

8.1 Heredity and Variation

Heredity is the transmission of characters (traits) from parents to offspring. **Variation** is the difference in traits among individuals of a species. Asexual reproduction produces little variation (only small DNA copying errors); sexual reproduction mixes two sets of DNA and creates more. Variations are the raw material that helps a population survive changing conditions.

Basic term	Meaning
Trait	A feature such as height, flower colour, seed shape
Gene	A segment of DNA that carries information for a protein and decides a trait
Allele	Alternative forms of a gene; one comes from each parent
Dominant trait	Expressed even if only one copy is present
Recessive trait	Expressed only when both copies are recessive
Genotype / phenotype	The genetic make-up / the visible appearance
Homozygous / heterozygous	Two identical alleles (TT, tt) / two different alleles (Tt)

8.2 Mendel's Contributions

Gregor Johann Mendel (1822 to 1884), an Austrian monk, performed careful experiments on **garden pea** (*Pisum sativum*) in the 1850s and 1860s. He chose peas because they have **contrasting traits** that are easy to see, a short life cycle, produce many seeds, and can be both self and cross pollinated.

Trait	Dominant	Recessive
Plant height	Tall (T)	Dwarf (t)
Seed shape	Round (R)	Wrinkled (r)
Seed colour	Yellow (Y)	Green (y)
Flower colour	Violet	White

▸ Monohybrid cross: one trait

A pure tall plant (TT) was crossed with a pure dwarf plant (tt). All **F_1** plants were **tall (Tt)**: the trait dwarfness did not blend. When F_1 plants were self-pollinated, the **F_2** generation had both tall and dwarf plants in a ratio of **3 : 1** (phenotype), and **1 TT : 2 Tt : 1 tt** (genotype).

	T	t
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (dwarf)

FORMULA

Monohybrid F_2 ratio: phenotypic **3 : 1**, genotypic **1 : 2 : 1**. **Dihybrid F_2 ratio (9 : 3 : 3 : 1):** 9 round-yellow : 3 round-green : 3 wrinkled-yellow : 1 wrinkled-green.

▸ Dihybrid cross: two traits

Mendel crossed round yellow seeds (RRYY) with wrinkled green seeds (rryy). F_1 were all **round yellow (RrYy)**, showing that round and yellow are dominant. F_1 self-pollination gave four types in F_2 in the ratio **9 : 3 : 3 : 1**. New combinations (round green, wrinkled yellow) appeared, so traits are inherited **independently** of each other.

DEFINITION

Law of segregation: the two alleles of a gene separate during gamete formation and each gamete gets one. **Law of independent assortment:** genes for different traits are inherited independently of one another.

8.3 How are Traits Expressed?

- Cells make **proteins** according to the DNA instructions; a protein, such as an enzyme or a hormone, controls a trait. For example, a plant-growth hormone enzyme decides whether a plant is tall; less of it gives a dwarf plant.
- Every individual receives **two copies** of each chromosome, one from each parent, so the equal contribution of each parent is seen in the offspring.
- **Chromosomes** in the nucleus carry genes; human cells have **23 pairs** (46).

8.4 Sex Determination

In humans, 22 pairs are **autosomes**; the 23rd pair is the **sex chromosomes**: **XX** in females and **XY** in males. All eggs carry X; sperm carry either X or Y, so the **father's sperm decides the sex of the child**.

	X (mother)	X (mother)
X (father)	XX girl	XX girl
Y (father)	XY boy	XY boy

- The chance of each child being a boy or a girl is **50 per cent**, independent of earlier children.
- In some animals (some reptiles) sex is decided by the temperature at which the eggs are incubated; in snails the sex can change over a lifetime.

8.5 Worked Examples

Cross Tt x tt. Gametes: T, t and t, t. Offspring: 2 Tt (tall) : 2 tt (dwarf), a 1 : 1 ratio. **Cross Tt x Tt:** 3 tall : 1 dwarf. If 600 plants come from a Tt x Tt cross, about **450 are tall** and 150 dwarf.

A parent with blood-group genes Aa and another aa: offspring 50 per cent Aa, 50 per cent aa.

Why can a recessive trait skip a generation? In heterozygous F_1 (Tt) the dominant allele masks the recessive one; it reappears in F_2 when two recessive alleles combine.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is a gene?	A segment of DNA that carries instructions for a protein and so decides a trait.
State the F_2 ratio of a monohybrid cross.	3 : 1 phenotypic and 1 : 2 : 1 genotypic.
Why did Mendel choose pea plants?	Contrasting traits, short life cycle, many seeds, easy to cross.
Who decides the sex of the child?	The father: sperm carries X or Y.
A cross between RrYy plants gives how many types?	Four types in the ratio 9 : 3 : 3 : 1.
What is the difference between a dominant and recessive trait?	Dominant shows even with one allele; recessive only with two.
What is the number of chromosomes in a human cell?	46 (23 pairs).
How does sexual reproduction increase variation?	DNA from two parents is combined and shuffled.

★ KEY TAKEAWAYS

- **Heredity** passes traits; **variation** creates differences among individuals.
- **Mendel's monohybrid F_2 ratio is 3 : 1; dihybrid is 9 : 3 : 3 : 1.**
- Each parent gives **one allele**; the dominant one shows in a heterozygote.
- Human sex chromosomes: **female XX, male XY**; the sperm decides the baby's sex, 50 : 50.